

EmPay

Smart HR Management System

Hackathon Development Plan | 24-Hour Sprint

Mohak

Frontend + Testing

Manav

Full Stack Bridge

Vismay

Frontend UI

Kishan

Backend / DevOps

Tech Stack

React + Vite

Node +
Express

MongoDB

JWT Auth

Claude Haiku
4.5

Recharts

Tailwind CSS

Python
(payslip)

4

Team Members

24 hrs

Sprint Duration

4

Core Modules

4

Parallel Tracks

Tech Stack Recommendation

WHY MERN + Python? Your team already knows MERN — the fastest path to a working demo. Python handles payslip PDF generation cleanly via ReportLab. Claude Haiku 4.5 (via GitHub Education Pack) sits as a middleware AI layer that auto-fills forms, summarises payroll, and generates HR notes.

Layer	Choice	Why	Who owns it
Frontend Framework	React 18 + Vite	Fast HMR, component reuse, familiar to team	Mohak + Vismay
UI Library	Tailwind CSS + shadcn/ui	Rapid styling, pre-built accessible components	Mohak + Vismay
Charts	Recharts	React-native, easy bar/pie/line charts for analytics	Vismay
Backend	Node.js + Express	MERN-native, REST APIs, middleware ecosystem	Kishan + Manav
Database	MongoDB Atlas (free tier)	Schema-flexible, cloud-hosted, no setup friction	Kishan
Auth	JWT + bcrypt	Stateless, role-payload in token, no session store needed	Kishan
AI Layer	Claude Haiku 4.5 (Anthropic API)	Form auto-fill, payroll summaries, leave reason generator	Manav
Payslip PDF	Python + ReportLab (microservice)	Professional PDF output, easy templating, REST endpoint	Kishan
State Mgmt	Zustand	Minimal boilerplate vs Redux, perfect for 24hr sprint	Manav
Form Handling	React Hook Form + Zod	Validation without re-renders, Zod schema matches backend	Mohak + Vismay
API Client	Axios + React Query	Caching, loading states, error handling out-of-box	Manav
Version Control	GitHub (meaningful commits)	Required deliverable, branch per module	All

CLAUDE HAIKU INTEGRATION POINTS: (1) Leave application — AI suggests reason text based on leave type. (2) Payroll summary — AI narrates the monthly payroll in plain English. (3) HR dashboard — AI insight card ('3 employees have >80% attendance this month'). (4) Payslip generation prompt — AI fills salary structure from employee profile. Keep prompts under 500 tokens so Haiku stays fast.

Project File Structure

Monorepo layout: /empay-root contains /client (React), /server (Node/Express), /pdf-service (Python), and /docs. Each member owns a folder — no merge conflicts on day 1.

```
empay/ ■■■ client/ # Mohak + Vismay (React + Vite) ■■■ public/ ■■■ src/ ■■■ api/ ■■■ ■■■
authApi.js # Login, register calls ■■■ ■■■ attendanceApi.js # Mark attendance, fetch logs ■■■ ■■■
leaveApi.js # Apply, approve, reject ■■■ ■■■ payrollApi.js # Payrun, payslip fetch ■■■ ■■■
employeeApi.js # CRUD employees ■■■ ■■■ aiApi.js # Claude Haiku wrapper (Manav) ■■■ ■■■ components/ ■■■
■■■ ui/ # shadcn primitives (Button, Card, Modal...) ■■■ ■■■ layout/ ■■■ ■■■ Sidebar.jsx #
Role-aware nav ■■■ ■■■ Navbar.jsx ■■■ ■■■ ProtectedRoute.jsx ■■■ ■■■ attendance/ ■■■ ■■■
■■■■ MarkAttendance.jsx ■■■ ■■■ AttendanceLogs.jsx ■■■ ■■■ leave/ ■■■ ■■■ LeaveApplyForm.jsx
■■■■ ■■■ LeaveStatus.jsx ■■■ ■■■ LeaveApprovalTable.jsx ■■■ ■■■ payroll/ ■■■ ■■■
PayrunList.jsx ■■■ ■■■ PayslipView.jsx ■■■ ■■■ SalaryBreakdown.jsx ■■■ ■■■ dashboard/ ■■■ ■■■
■■■■ AdminDashboard.jsx ■■■ ■■■ EmployeeDashboard.jsx ■■■ ■■■ HRDashboard.jsx ■■■ ■■■
PayrollDashboard.jsx ■■■ ■■■ employees/ ■■■ ■■■ EmployeeList.jsx ■■■ ■■■ EmployeeForm.jsx ■■■ ■■■
■■■■ EmployeeProfile.jsx ■■■ ■■■ pages/ ■■■ ■■■ Login.jsx ■■■ ■■■ Register.jsx ■■■ ■■■
Dashboard.jsx # Routes to role-specific dashboard ■■■ ■■■ AttendancePage.jsx ■■■ ■■■ LeavePage.jsx ■■■
■■■■ PayrollPage.jsx ■■■ ■■■ EmployeesPage.jsx ■■■ ■■■ ProfilePage.jsx ■■■ ■■■ store/ ■■■ ■■■
■■■■ authStore.js # Zustand: user + role + token ■■■ ■■■ notifStore.js # Toast notifications ■■■ ■■■
utils/ ■■■ ■■■ payrollCalc.js # PF(12%), Prof Tax, net pay logic ■■■ ■■■ dateHelpers.js ■■■ ■■■
roleGuard.js # Route permission map ■■■ ■■■ App.jsx ■■■ ■■■ main.jsx ■■■ ■■■ .env # VITE_API_URL,
VITE_AI_ENABLED ■■■ ■■■ package.json ■■■ ■■■ server/ # Kishan + Manav (Node/Express) ■■■ ■■■ config/ ■■■ ■■■
db.js # Mongoose connect ■■■ ■■■ env.js # dotenv loader ■■■ ■■■ models/ ■■■ ■■■ User.js # name, email,
password, role, employeeId ■■■ ■■■ Employee.js # dept, designation, wage, joinDate, pfOpt ■■■ ■■■
Attendance.js # employeeId, date, checkIn, checkOut, status ■■■ ■■■ Leave.js # employeeId, type, dates,
status, approvedBy ■■■ ■■■ LeaveBalance.js # employeeId, annual/sick/casual remaining ■■■ ■■■ Payrun.js #
month, year, status, createdBy ■■■ ■■■ Payslip.js # payrunId, employeeId, earnings, deductions, net ■■■ ■■■
routes/ ■■■ ■■■ auth.routes.js # POST /register, /login, /me ■■■ ■■■ employee.routes.js # CRUD /employees
■■■■ ■■■ attendance.routes.js # POST /mark, GET /logs, GET /stats ■■■ ■■■ leave.routes.js # Apply,
approve, reject, list ■■■ ■■■ payroll.routes.js # Payrun CRUD, payslip generate ■■■ ■■■ ai.routes.js #
POST /ai/suggest (Haiku proxy) ■■■ ■■■ controllers/ ■■■ ■■■ auth.controller.js ■■■ ■■■
employee.controller.js ■■■ ■■■ attendance.controller.js ■■■ ■■■ leave.controller.js ■■■ ■■■
payroll.controller.js # payroll calc engine lives here ■■■ ■■■ ai.controller.js # calls Anthropic API ■■■
■■■■ middleware/ ■■■ ■■■ auth.middleware.js # verifyToken ■■■ ■■■ role.middleware.js #
requireRole('admin','hr',...) ■■■ ■■■ utils/ ■■■ ■■■ payrollEngine.js # ACCURATE calc: basic, HRA, PF, prof
tax, net ■■■ ■■■ .env # MONGO_URI, JWT_SECRET, ANTHROPIC_API_KEY ■■■ ■■■ server.js ■■■ ■■■ package.json ■■■ ■■■
pdf-service/ # Kishan (Python microservice) ■■■ ■■■ main.py # FastAPI app ■■■ ■■■ templates/ ■■■ ■■■
payslip_template.py # ReportLab payslip renderer ■■■ ■■■ requirements.txt # fastapi, uvicorn, reportlab ■■■
■■■■ .env ■■■ ■■■ docs/ ■■■ ■■■ API.md # Endpoint reference ■■■ ■■■ SETUP.md # Run instructions ■■■ ■■■
PAYROLL_LOGIC.md # Formula documentation
```

Payroll Calculation Logic (Accurate)

CRITICAL: All payroll numbers must be derived from these formulas. Implement payrollEngine.js on the backend — never compute salary on the frontend.

```
// server/utls/payrollEngine.js //
// INPUTS (from Employee +
Attendance + Leave models) // wage : Monthly CTC (from Employee.wage) // totalDays : Working days in the
month (e.g. 26) // presentDays : Days actually present (attendance records) // approvedLeaveDays : Approved
paid leaves (Leave model) // pfOptIn : Boolean – employee opted into PF? // // STEP 1 – Paid days //
paidDays = presentDays + approvedLeaveDays // // STEP 2 – Gross Salary (pro-rated) // grossSalary = (wage /
totalDays) * paidDays // // STEP 3 – Salary Components (% of grossSalary) // basic = grossSalary * 0.50 //
50% // hra = grossSalary * 0.20 // 20% // special = grossSalary * 0.30 // 30% (special allowance) // // STEP
4 – Deductions // pfEmployee = pfOptIn ? basic * 0.12 : 0 // 12% of basic // pfEmployer = pfOptIn ? basic *
0.12 : 0 // shown in CTC, not deducted // profTax = getProfTax(grossSalary) // state slab // // STEP 5 – Net
Pay // netPay = grossSalary - pfEmployee - profTax // // PROFESSIONAL TAX SLABS (Maharashtra) function
getProfTax(gross) { if (gross <= 7500) return 0; if (gross <= 10000) return 175; return 200; // above
10,000/month } // // PAYSHEET FIELDS TO DISPLAY // Earnings : Basic, HRA, Special Allowance, Gross //
Deductions: PF (Employee), Professional Tax, Total Deductions // Net Pay = Gross - Total Deductions //
```

Sample Payslip Verification

Component	Formula	Example (CTC 50,000, 22/26 days, PF on)
Gross Salary	$(50000/26)*22$	Rs. 42,307.69
Basic (50%)	$42307.69 * 0.50$	Rs. 21,153.85
HRA (20%)	$42307.69 * 0.20$	Rs. 8,461.54
Special Allowance (30%)	$42307.69 * 0.30$	Rs. 12,692.31
PF Deduction (12% basic)	$21153.85 * 0.12$	- Rs. 2,538.46
Professional Tax	Maharashtra slab	- Rs. 200.00
NET PAY	Gross - PF - Prof Tax	Rs. 39,569.23

Work Distribution by Member

Mohak — Frontend + Basic Testing

Timeline

0-8: Auth + Layout | 8-16: Attendance + Leave UI | 16-22: Profile + Testing | 22-24: Bug fixes

Task	Description
Auth UI	Login.jsx, Register.jsx — role-aware redirect after login
Layout	Sidebar.jsx (role-based nav items), Navbar.jsx, ProtectedRoute.jsx
Profile Page	ProfilePage.jsx — view & edit own profile, avatar upload
Attendance UI	MarkAttendance.jsx (check-in/out button + timer), AttendanceLogs.jsx (calendar heatmap)
Leave Apply	LeaveApplyForm.jsx — type selector, date range picker, AI-suggested reason (Haiku)
Leave Status	LeaveStatus.jsx — employee view of own leave requests + status badges
Testing	Manual test checklist: auth flows, leave apply, attendance mark, payslip view

Files Owned

src/pages/Login.jsx | src/pages/Register.jsx | src/components/layout/Sidebar.jsx | src/components/layout/Navbar.jsx | src/components/layout/ProtectedRoute.jsx | src/pages/ProfilePage.jsx | src/components/attendance/MarkAttendance.jsx | src/components/attendance/AttendanceLogs.jsx ... +2 more

Claude Haiku Task for Mohak: LeaveApplyForm — POST /ai/suggest with leave type → Haiku returns a professional reason string. Show it as a 'Suggested reason' chip that employee can accept or edit.

Manav — Full Stack Bridge

Timeline

0-6: API layer + Zustand | 6-14: AI controller + hooks | 14-20: PayrunList + integration | 20-24: Polish + bug fix

Task	Description
API Layer (client)	Create all src/api/*.js files — centralised Axios calls with token injection
Zustand Stores	authStore.js (user, role, token, login/logout actions), notifStore.js
React Query Setup	queryClient config, useAttendance, useLeave, usePayroll custom hooks
AI Route (server)	ai.routes.js + ai.controller.js — proxies Anthropic Haiku calls server-side
AI UI Component	Floating 'Ask HR AI' button on dashboards — summarises stats, answers queries
Payroll UI Bridge	PayrunList.jsx — connects backend payrun data to table, triggers PDF download
Integration Testing	Verify full flow: Employee marks attendance → HR views it → Payroll Officer runs payrun → Payslip appears

Files Owned

src/api/authApi.js | src/api/attendanceApi.js | src/api/leaveApi.js | src/api/payrollApi.js |
src/api/employeeApi.js | src/api/aiApi.js | src/store/authStore.js | src/store/notifStore.js ... +6 more

Claude Haiku Task for Manav: ai.controller.js calls Anthropic /v1/messages with model 'claude-haiku-4-5'. Prompts: (a) Leave reason helper — max 100 tokens. (b) Payroll narrative — 'Summarise this month payroll data in 2 sentences' — max 150 tokens. (c) Dashboard insight — 'Which employees had <70% attendance?' from JSON data — max 200 tokens.

Vismay — Frontend UI / Dashboards

Timeline

0-6: Dashboard shells + routing | 6-14: Recharts integration + data binding | 14-20: Employee pages + leave table | 20-24: Payslip view + polish

Task	Description
Admin Dashboard	AdminDashboard.jsx — total employees, attendance rate, pending leaves, payroll summary cards + bar chart
HR Dashboard	HRDashboard.jsx — attendance overview table (all employees), leave pending count, employee directory link
Payroll Dashboard	PayrollDashboard.jsx — payrun status, total payroll amount, payslip generation button
Employee Dashboard	EmployeeDashboard.jsx — own attendance %, leave balance, last payslip preview
Analytics Charts	Recharts: Monthly attendance bar chart, leave type pie chart, department-wise payroll bar chart
Employee Pages	EmployeeList.jsx (searchable table), EmployeeForm.jsx (create/edit), EmployeeProfile.jsx
Leave Approval Table	LeaveApprovalTable.jsx — for HR Officer & Payroll Officer, approve/reject buttons
Payslip View	PayslipView.jsx — formatted payslip display with all salary components

Files Owned

src/components/dashboard/AdminDashboard.jsx | src/components/dashboard/HRDashboard.jsx |
src/components/dashboard/PayrollDashboard.jsx | src/components/dashboard/EmployeeDashboard.jsx |
src/components/employees/EmployeeList.jsx | src/components/employees/EmployeeForm.jsx |
src/components/employees/EmployeeProfile.jsx | src/components/leave/LeaveApprovalTable.jsx ... +3 more

Claude Haiku Task for Vismay: AdminDashboard AI Insight Card — fetch last month attendance stats, send as JSON to /ai/suggest endpoint, display Haiku's 1-2 sentence insight in a highlighted card (e.g. 'Attendance dropped 12% vs last month in Engineering dept.')

Kishan — Backend + DevOps

Timeline

0-4: Repo + DB setup + Models | 4-10: Auth + Employee APIs | 10-16: Attendance + Leave APIs | 16-21: Payroll engine + PDF service | 21-24: Integration fixes + deploy

Task	Description
DB Models	User, Employee, Attendance, Leave, LeaveBalance, Payrun, Payslip — with indexes on employeeid + date
Auth APIs	POST /api/auth/register, /login, /me — JWT with role in payload, bcrypt password hashing

Task	Description
Employee APIs	CRUD /api/employees — only Admin & HR Officer can create/edit, all roles can read
Attendance APIs	POST /api/attendance/mark (idempotent per day), GET /logs?empld&month;, GET /stats (for payroll engine)
Leave APIs	POST /apply, PATCH /approve/:id, PATCH /reject/:id, GET /list — with role checks
Payroll Engine	payrollEngine.js — accurate PF, professional tax, pro-rated salary (see formulas page)
Payroll APIs	POST /api/payroll/payrun (triggers engine for all employees), GET /payslip/:id
Python PDF Service	FastAPI + ReportLab — POST /generate-payslip → returns PDF binary; called from Node payroll controller
MongoDB Atlas	Free cluster setup, .env config, connection pooling, basic indexes
GitHub Setup	Init repo, branch strategy (main/dev/feature/*), protect main branch

Files Owned

```
server/models/User.js | server/models/Employee.js | server/models/Attendance.js | server/models/Leave.js |
server/models/LeaveBalance.js | server/models/Payrun.js | server/models/Payslip.js |
server/routes/auth.routes.js ... +16 more
```

Claude Haiku Task for Kishan: No direct Haiku integration — Kishan exposes the /api/ai route (Manav builds the controller). Kishan ensures ANTHROPIC_API_KEY is in .env and the route is secured with verifyToken middleware.

24-Hour Sprint Timeline

Hour 0-2	SETUP Kishan: init repo + MongoDB Atlas + .env. Manav: Vite project + Tailwind + folder structure. Mohak & Vismay: clone repo, install deps, confirm dev server runs.
Hour 2-8	BACKEND CORE Kishan: User/Employee/Attendance/Leave models + Auth APIs. Manav: src/api/ files (stubs, token interceptor), authStore. All can now call /register & /login.
Hour 8-14	FEATURE BUILD Kishan: Attendance + Leave APIs + payroll engine. Manav: AI controller + React Query hooks. Mohak: Auth pages + Layout + Attendance UI. Vismay: Dashboard shells + charts.
Hour 14-20	INTEGRATION Kishan: Payroll APIs + Python PDF service. Manav: Connect all hooks to UI, payrun flow end-to-end. Mohak: Leave apply + status pages. Vismay: Employee pages + leave approval table.
Hour 20-22	POLISH + AI Manav: Wire Haiku into leave form + dashboard insight card. Vismay: Payslip view + salary breakdown. Mohak: Profile page + manual test run. Kishan: Fix any API bugs, check payroll accuracy.
Hour 22-24	SUBMISSION All: Final manual smoke test. Kishan: Tag v1.0 on GitHub with meaningful commit messages. Mohak: Write SETUP.md. Manav: Demo walkthrough script.

Git Branch Strategy

Branch	Owner	Merge into
main	All	— (protected)
dev	All	main at submission
feature/auth	Kishan	dev
feature/attendance	Kishan + Mohak	dev
feature/leave	Kishan + Mohak	dev
feature/payroll	Kishan + Manav	dev
feature/dashboard	Vismay	dev
feature/ai	Manav	dev

API Endpoint Reference

Auth

Method	Endpoint	Notes	Response
POST	/api/auth/register	Admin only. Body: {name, email, password, role}	201 {user}
POST	/api/auth/login	Body: {email, password}	200 {token, user}
GET	/api/auth/me	Bearer token required	200 {user}

Employees

Method	Endpoint	Notes	Response
GET	/api/employees	All roles. Query: ?dept, ?search	200 [{employee}]
POST	/api/employees	Admin, HR only. Body: employee object	201 {employee}
PATCH	/api/employees/:id	Admin, HR only	200 {employee}
DELETE	/api/employees/:id	Admin only	204

Attendance

Method	Endpoint	Notes	Response
POST	/api/attendance/mark	Employee. Body: {type: 'checkin' 'checkout'} — idempotent per day	200 {record}
GET	/api/attendance/logs	Query: ?empId&month;&year.; Role-filtered	200 [{log}]
GET	/api/attendance/stats	Payroll Officer. Returns presentDays per employee per month	200 {stats}

Leave

Method	Endpoint	Notes	Response
POST	/api/leave/apply	Employee. Body: {type, startDate, endDate, reason}	201 {leave}
GET	/api/leave/list	Role-filtered: employee sees own, HR/Payroll sees all	200 [{leave}]
PATCH	/api/leave/approve/:id	Payroll Officer only	200 {leave}
PATCH	/api/leave/reject/:id	Payroll Officer only. Body: {reason}	200 {leave}
GET	/api/leave/balance/:empId	Employee sees own, HR sees all	200 {balance}

Payroll

Method	Endpoint	Notes	Response
POST	/api/payroll/payrun	Payroll Officer only. Body: {month, year} — runs engine for all employees	201 {payrun}

Method	Endpoint	Notes	Response
GET	/api/payroll/payrun	List all payruns. Payroll Officer / Admin	200 [{payrun}]
GET	/api/payroll/payslip/:empId	Employee sees own; Payroll Officer sees all	200 {payslip}
GET	/api/payroll/payslip/:id/pdf	Calls Python service, returns PDF	200 PDF binary

AI (Haiku)

Method	Endpoint	Notes	Response
POST	/api/ai/suggest	Any role. Body: {type: 'leave' 'insight' 'summary', data: {}}	200 {suggestion}

Tips, Pitfalls & Haiku Prompts

NEVER compute payroll on frontend

payrollEngine.js runs server-side only. Frontend only displays what the API returns. This prevents rounding errors and ensures a single source of truth.

Idempotent attendance marking

One attendance record per employee per day. Use findOneAndUpdate with upsert on {employeeId, date}. Calling mark twice should update, not create duplicate.

JWT role in payload

Include role in JWT payload: jwt.sign({userId, role}, secret). Middleware extracts it — no extra DB call per request.

MongoDB Atlas free tier limits

512 MB storage. Keep attendance records lean — only store {empId, date, checkIn, checkOut, status}. No unnecessary fields.

React Query cache keys

Use consistent keys: ['attendance', empId, month]. Invalidate after mark attendance so logs refresh automatically.

Payroll is idempotent too

Running payroll twice for same month should update, not duplicate. Use upsert on {payrunId, employeeId}.

Python service on different port

Run pdf-service on :8001, Node on :5000, React on :3000. Add proxy in vite.config.js for /api → :5000.

Ready-to-Use Claude Haiku Prompts

Leave Reason Helper

System: "You are an HR assistant. Return only a professional leave reason in 1-2 sentences." User: "Employee is taking Sick Leave from {startDate} to {endDate}. Suggest a formal reason."

Payroll Narrative

System: "You are a payroll analyst. Return exactly 2 sentences summarising the data." User: "Month: {month}. Total employees: {count}. Total payout: Rs.{total}. Avg net pay: Rs.{avg}. Top earner dept: {dept}."

Attendance Insight

System: "You are an HR analytics bot. Return 1 insight sentence only." User: "Attendance data JSON: {json}. Identify any department with attendance below 75%."

Payslip Structure Helper

System: "Return JSON only: {basic, hra, special, pf, profTax, net}. No explanation." User: "Employee CTC: {wage}, present days: {present}, total days: {total}, PF opted: {pfOptIn}."